SECURITY ENGINEERING RESEARCH POSTER

Measuring Neural-Network Robustness Under Adversarial Perturbation

Experimental Evaluation of Model Behavior Under Gradient-Based Attacks

github.com/poojakira/adversarial-ml-lab - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

112	32.22%	1,117,354	71.82%
Tests passed	Statement coverage	SmallCNN parameters	Clean CIFAR-10 accuracy

SYSTEM VIEW

Architecture and Threat-Control Flow

Trust Boundary

Input surface
Data, models, requests, policies, or artifacts entering the tested path.

Control surface
Repository-specific validation, analysis, and fail-closed checks.

Evidence surface
Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01 Train/evaluate a compact CIFAR-10 classifier under a fixed seed and CPU budget. →

02 Measure clean accuracy on the full CIFAR-10 test set. ↓

03 Generate epsilon-bounded FGSM and PGD adversarial samples plus C&W L2 samples. →

04 Measure robust accuracy on the committed attack subset. ↓

05 Emit structured evidence and CI robustness-gate logic.

Research Contribution

A reproducible robustness-evaluation harness for FGSM, PGD, and C&W that shows a measured 71.82% clean CIFAR-10 SmallCNN collapsing to 0.00% robust accuracy under PGD-20 at epsilon 8/255, while explicitly separating measured results from literature projections.

Verified Evidence

EV-01	FGSM robust accuracy @ 8/255: 3.32%	EV-02	PGD-20 robust accuracy @ 8/255: 0.00%
EV-03	C&W L2 robust accuracy: 4.20%	EV-04	Robust attack evaluation subset: 1,024 samples
EV-05	112 tests passed.	EV-06	32.22% statement coverage; CI gate is 15%.
EV-07	Lint/format, security audit, and CodeQL jobs succeeded.	EV-08	The current CI robustness benchmark job is skipped on ordinary pushes, so the benchmark values below come from the committed real artifact rather than this CI run.

Method and Control Path

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Limitations and Residual Risk

- This repository measures vulnerability; it does not make a model robust.
- Robust-accuracy attacks use a 1,024-sample subset, not the full 10,000 test set.
- No AutoAttack, black-box transfer, or certified-robustness result is established.
- results/cifar10_resnet18_benchmark.json is explicitly literature-projected/synthetic and must not be presented as measured.

Reproducibility

```
Expected evidence at the cited snapshot: 112 passed,
32.22% coverage. Measured code snapshot:
05ee102c4c20fe3ebdcd3a8ac3dd1ea530de6698 Evidence CI run:
38498116492
```

Evidence Provenance

Claim discipline
Metrics come from the repository's verified metrics and research brief.

Scope discipline
Benchmarks are reported only at their documented fixture, dataset, or test scope.

Reproduction
PDF and PNG are generated locally from committed evidence without paid services.

Evidence-backed security engineering

README, verified metrics, tests, CI, and committed artifacts remain authoritative. Poster layout emphasizes architecture, controls, evidence, limitations, and reproduction rather than unsupported efficacy claims.

36 x 48 in
PDF + PNG parity